

Lecture 5

Measures of Location and Position

Review

1 A distribution has two clear peaks. What does that usually tell us?

The sample mixes two distinct sub-populations, picked out by some grouping variable.

2 What makes a distribution skewed?

A hard limit on the values – data pile up against the limit and trail away from it.

3 Name the two very different things an outlier can be.

A genuinely unusual observation, or a measurement or recording error.

4 Which plot handles a large sample best, and which one keeps the actual data values?

The histogram for large n . The stem and leaf plot keeps the original values.

Review: one small dataset

x = waiting time between eruptions of Old Faithful (minutes), for a random sample of $n = 10$ eruptions.

$x = \{ 45, 51, 52, 56, 77, 78, 78, 78, 79, 84 \}$

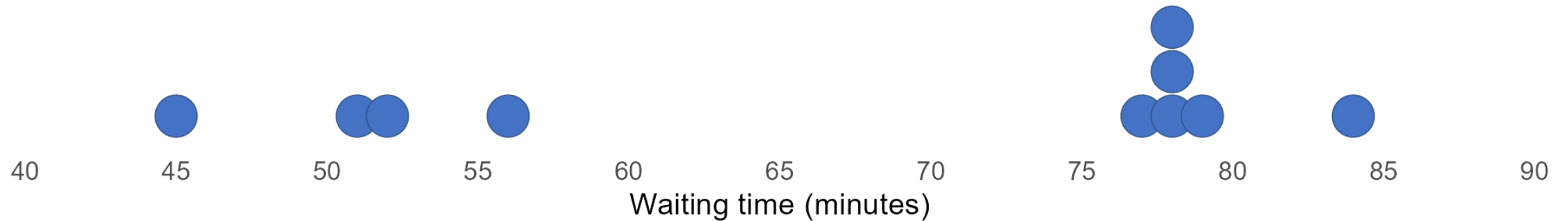
Waiting time (minutes)	Frequency	Relative Frequency
40 to under 50	1	0.1
50 to under 60	3	0.3
60 to under 70	0	0.0
70 to under 80	5	0.5
80 to under 90	1	0.1
Total	10	1.0

Five bins of width 10. Notice the bin that came out empty.

Your turn: the dot plot

Draw the dot plot. One dot per observation, stacked where a value repeats.

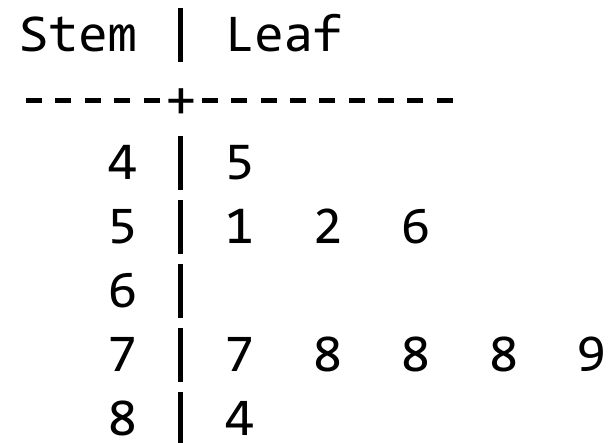
$x = \{ 45, 51, 52, 56, 77, 78, 78, 78, 79, 84 \}$



Your turn: the stem and leaf plot

Build the stem and leaf plot. Stems are the tens digit, leaves the ones.

x = { 45, 51, 52, 56, 77, 78, 78, 78, 79, 84 }

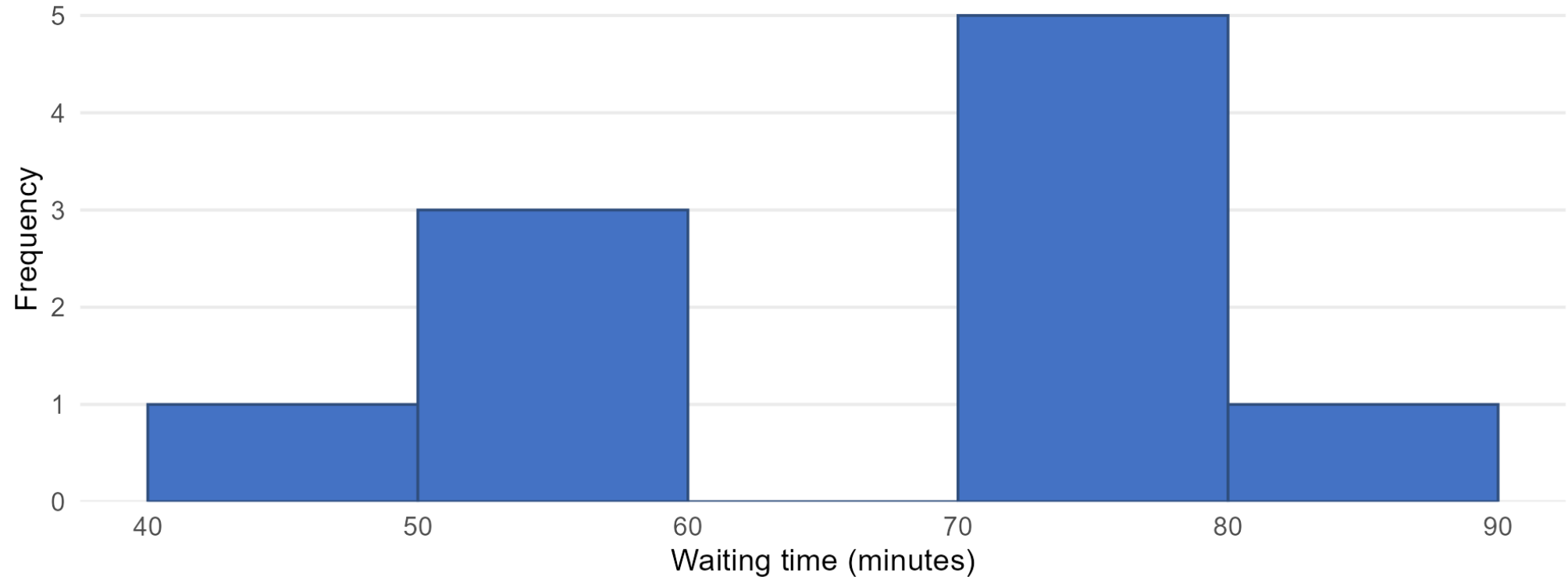


Key: 4 | 5 = 45 minutes

Your turn: the histogram

Draw the histogram using the five bins from the frequency table.

$x = \{ 45, 51, 52, 56, 77, 78, 78, 78, 79, 84 \}$



What the three plots agree on

- All three show the same gap: not one waiting time between 60 and 70 minutes
- Two clusters, around 50 minutes and around 78
 - The same two humps we saw on Wednesday in all 272 eruptions
- But with $n = 10$ the shape is faint. With $n = 272$ it was unmistakable
 - A small sample can hide a feature that is really there
- So eyeballing a picture only takes us so far. Now we compute numbers
 - Measures of location: where is the center?
 - Measures of position: where does one observation sit relative to the rest?

Measures of Central Tendency

- The (arithmetic) **mean** is the average value of a set of observations
it measures the center of mass of a distribution (the balancing point)

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \sum_{i=1}^n \frac{x_i}{n}$$

- the mean is usually not equal to any of the values observed in the sample
- The mean is highly influenced by **outliers** - observations that take on extreme values relative to the distribution

Practice: Calculate The Mean

- $X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

Measures of Central Tendency

- The **median** is the middle value of a set of observations

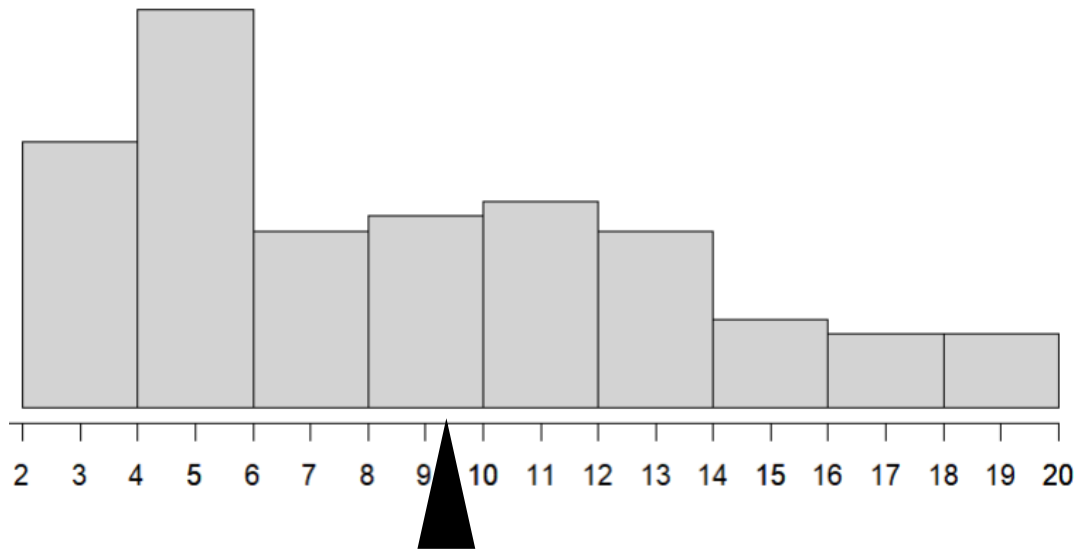
How to compute the median:

1. Compute the median by first ordering the observations from smallest value to largest value and choose the number in the middle
2. If the n is odd the median is the middle number
 - If n is even the median is the sum of the two middle values divided by 2

Practice: Calculate the Median

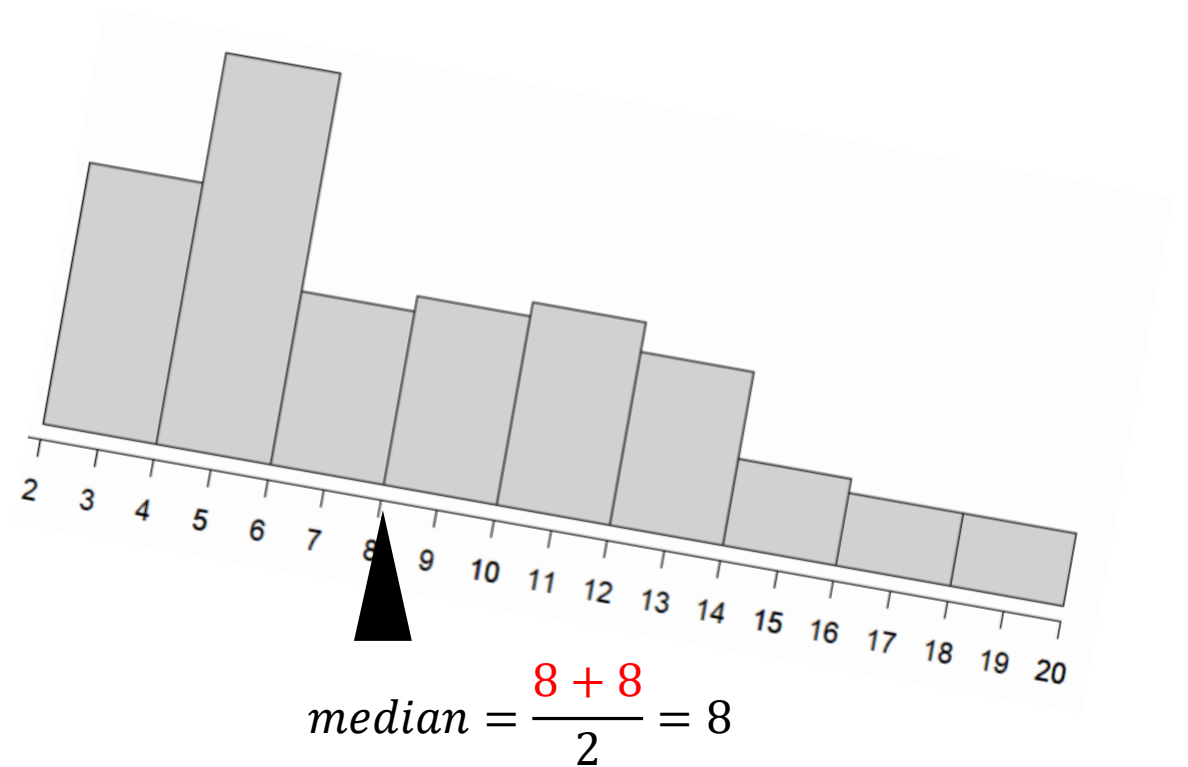
- $X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

Data: 3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 5 5 5 5 5 5 5 5 5 5
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$$\bar{x} = 9.2$$

The mean is the center of gravity

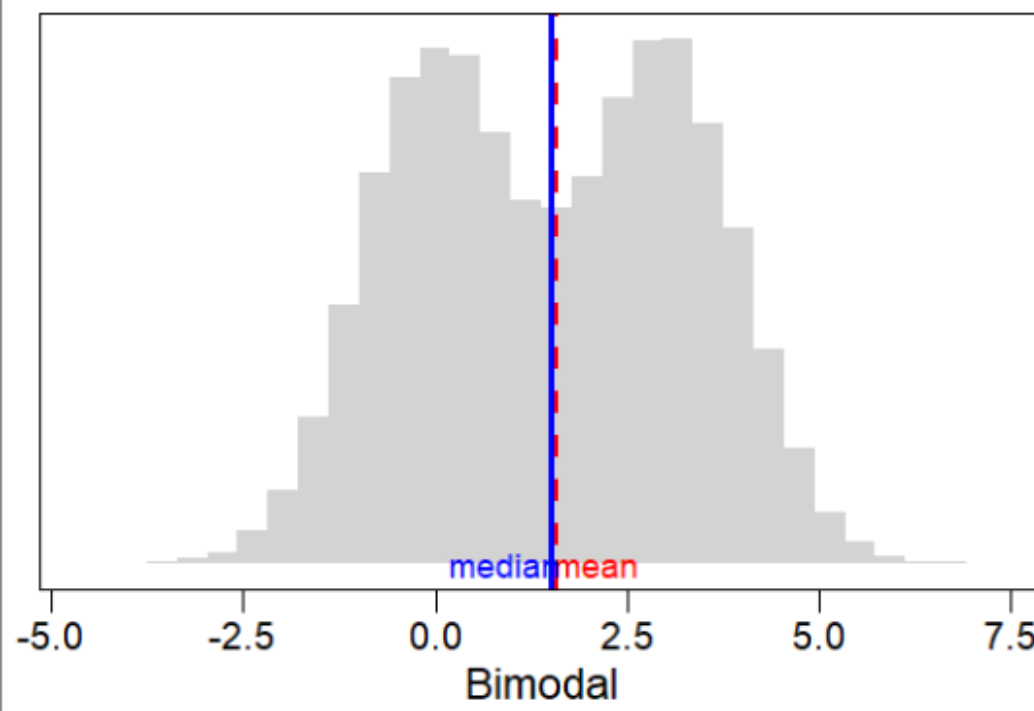
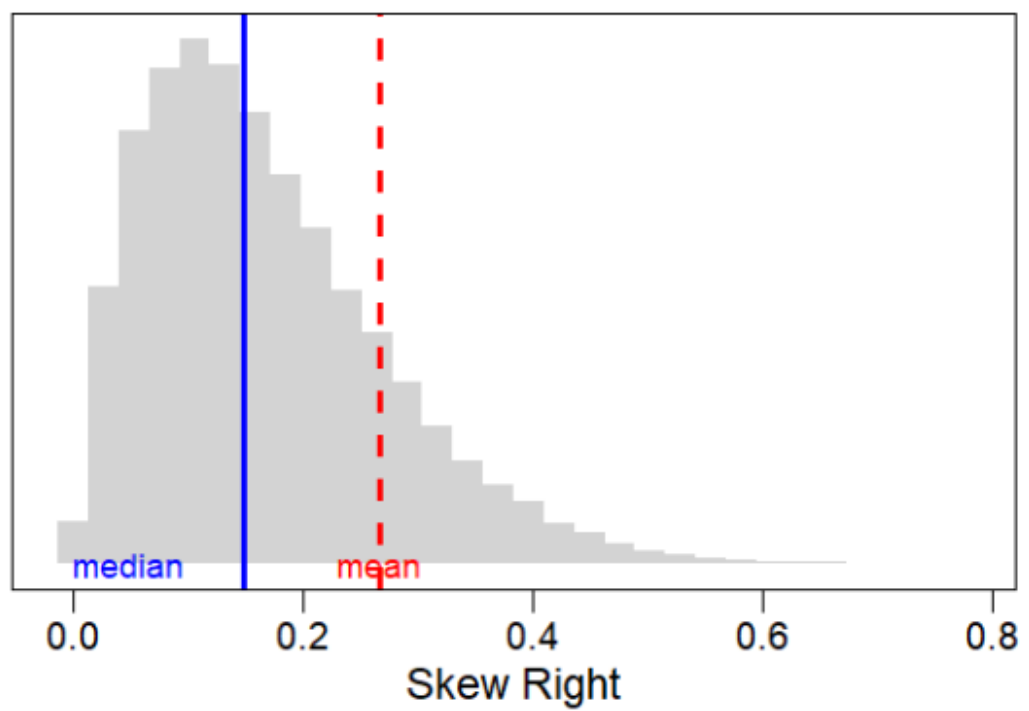
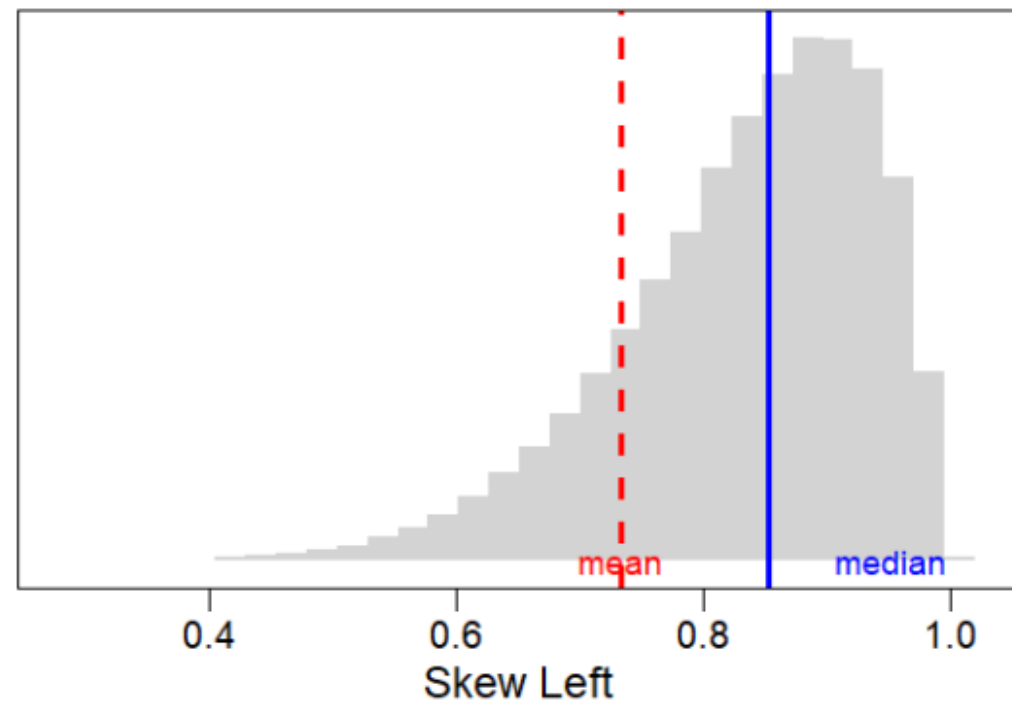
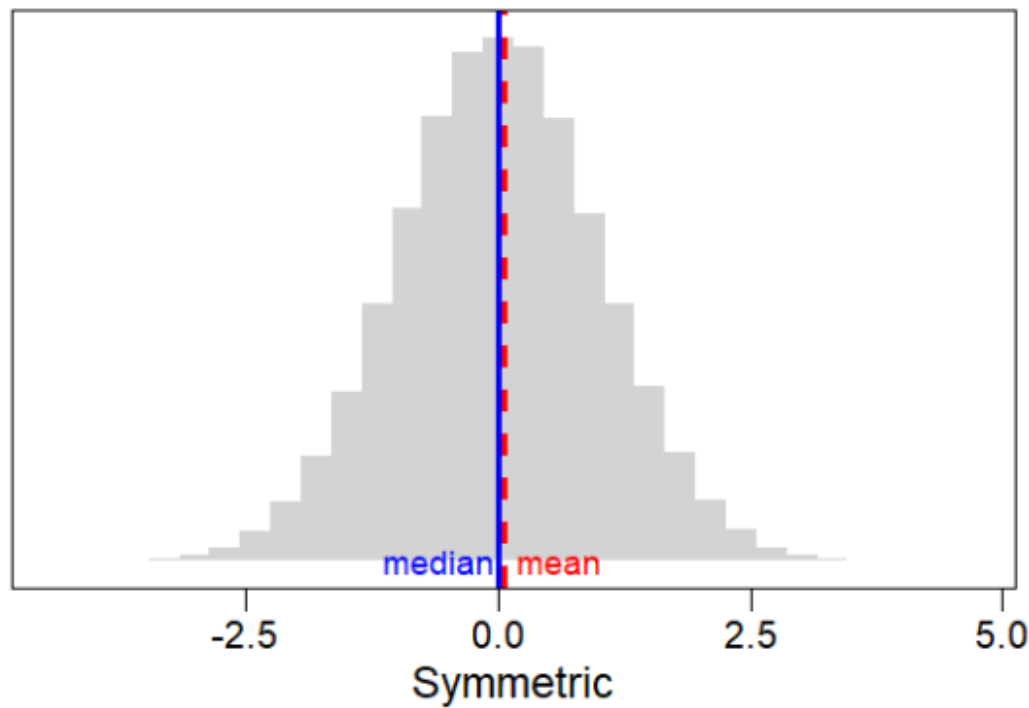


$$median = \frac{8 + 8}{2} = 8$$

The median is the middle value

Mean and median treat outliers differently

- $X = \{1, 3, 5, 5, 6, 7, 7, 8, 32\}$



Alternative formulas for the mean

- We can also express the mean in terms of the frequency F or the relative frequency RF

$$\bar{x} = \frac{1}{n} \sum_x x F(x) \quad \text{or} \quad \bar{x} = \sum_x x RF(x)$$

Where the sum is over all distinct values of the variable x

Example: Computing the mean from a frequency table

$X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

X	Freq.	Rel. Freq
1	1	0.125
3	1	0.125
5	2	0.250
6	1	0.125
7	2	0.250
8	1	0.125

The mode

- The **mode** is the value with the largest relative frequency (i.e the value that occurs most often)
 - Can be used with categorical data (mean and median cannot)
 - e.g the most frequent category
 - It may not be unique if two or more values have the same frequency
 - **Caution** for quantitative data, the mode may not anywhere near the center of the distribution.

Ex.)

Data = 1,1,4,5,6

Mode = 1

Data = 1,1,4,5,6,6

Mode 1, 6

Practice:

- Roll a six-sided die $n = 10$ times and record the number rolled each time
- Data = 1,2,3,3,4,4,4,5,6,6

x	$f(x)$	$rf(x)$
1	1	0.1
2	1	0.1
3	2	0.2
4	3	0.3
5	1	0.1
6	2	0.2

Compute the **mean** using all 3 equations:

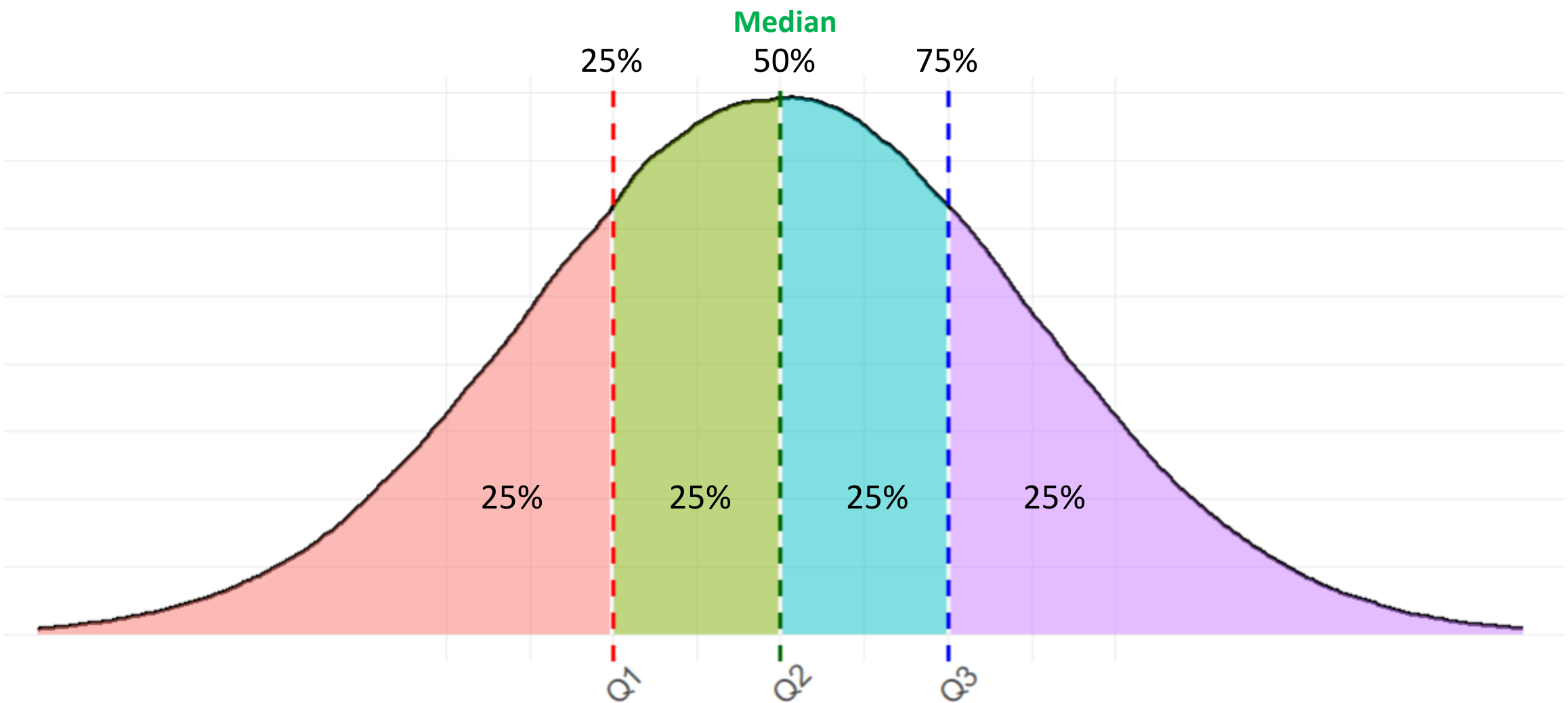
Compute the **median**

Compute the **mode**

Measures of Location: Quartiles and Percentiles

- The p^{th} **percentile** is a value such that p percent of the observations in a sample or population fall at or below that value
- Ex. The 50th percentile of any dataset is the median
- Three useful percentiles of a distribution are the **quartiles**
 1. The first **quartile Q1** is the 25th percentile of the data.
 2. The second **quartile Q2** is the 50th percentile or median of the data.
 3. The third **quartile Q3** is the 75th percentile of the data.
- The quartiles split a distribution in four equal parts each containing 25% of the observations

Measures of Position: Quartiles and Percentiles



Measures of Position: Quartiles and Percentiles

- How to compute the quartiles:
 1. Arrange the data in increasing order
 2. Find the median and label as **Q2**
 3. Consider the lower half of the observations (excluding the median itself if n is an odd number).
 4. Mark the median for the lower half of the observations and label as **Q1**
 5. Consider the upper half of the observations (again excluding the median itself if n is odd)
 6. Mark the median for the upper half of the observations and label as **Q3**

Ex. Quartiles

- Consider the following data which come from $n = 20$ rolls of a six-sided die

• Data =

lower half	middle	upper half
1, 1, 1, 2, 2, 2, 2, 2	2, 3	3, 3, 3, 3, 4, 5, 5, 6, 6, 6

$$Q2 = \text{median} = \frac{2+3}{2} = 2.5$$

$$Q1 = \text{median lower half} = \frac{2+2}{2} = 2$$

$$Q3 = \text{median upper half} = \frac{4+5}{2} = 4.5$$

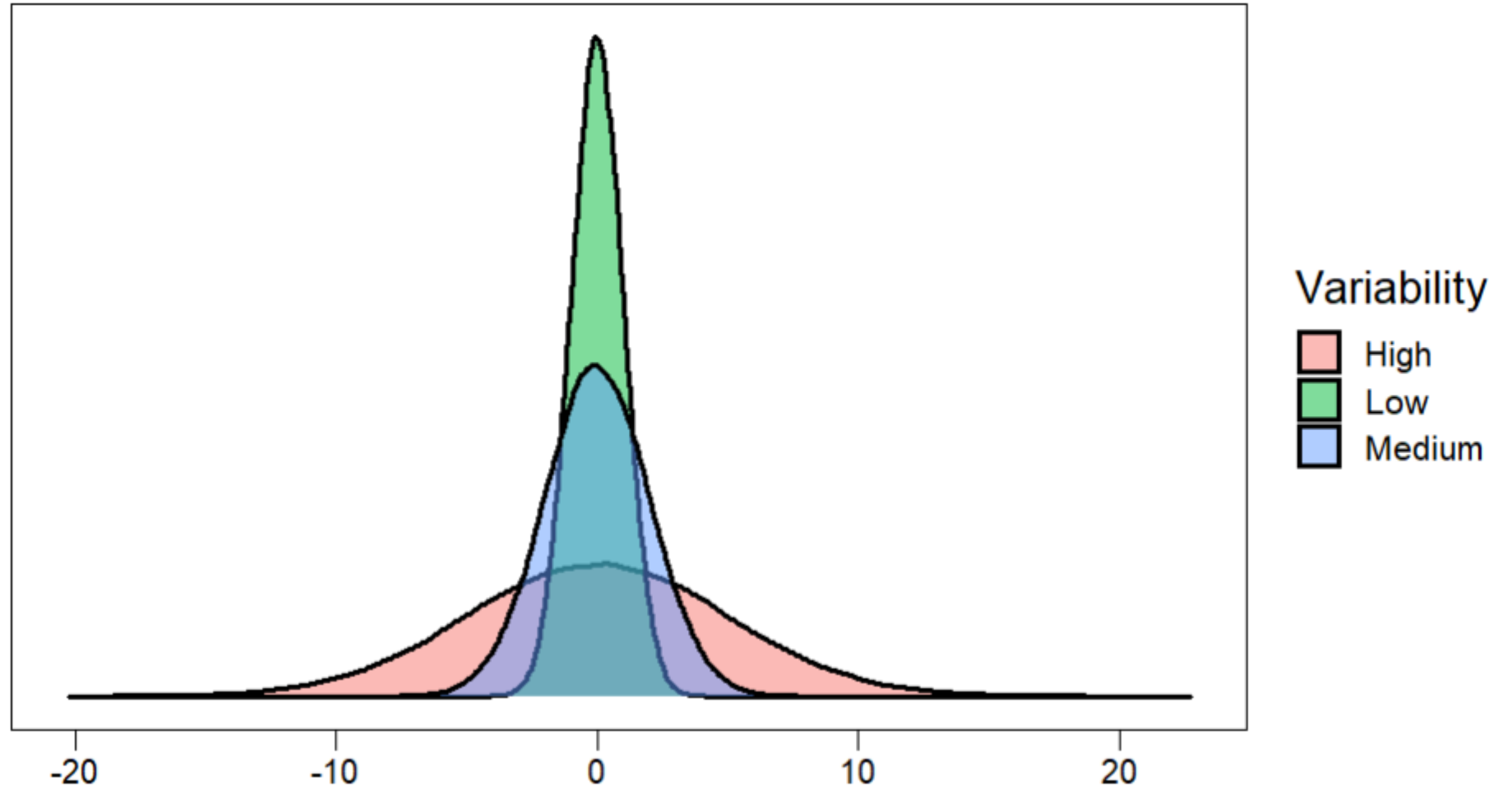
Practice

Consider the following 15 exam scores of students in a statistics course

61,61,65,65,66,68,69,73,74,75,76,78,79,90,94

Compute the 3 quartiles Q_1 , Q_2 , and Q_3

Variability of A Distribution: Measures of Spread

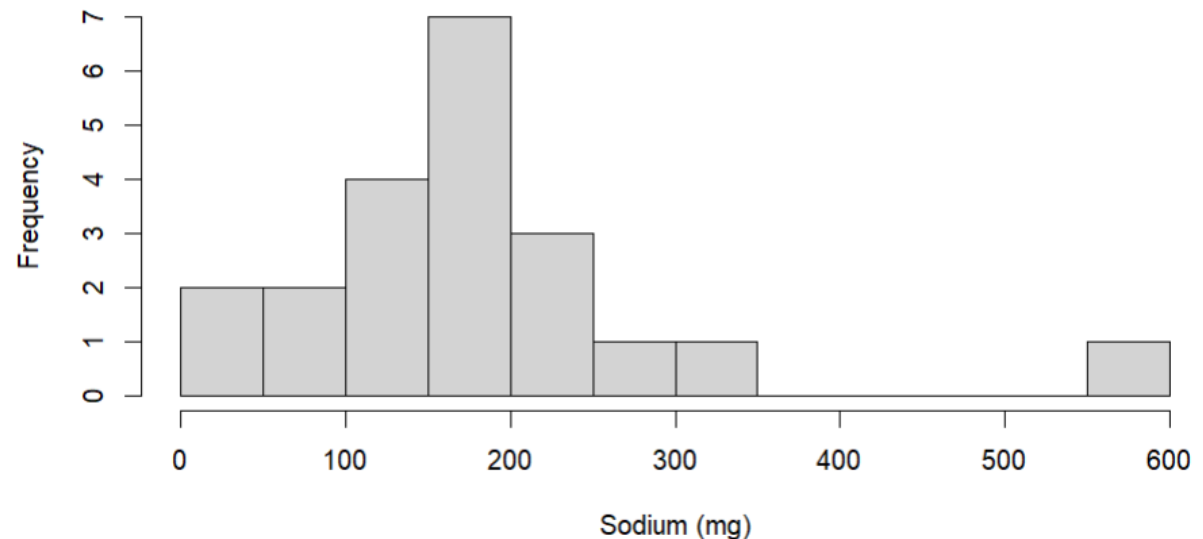
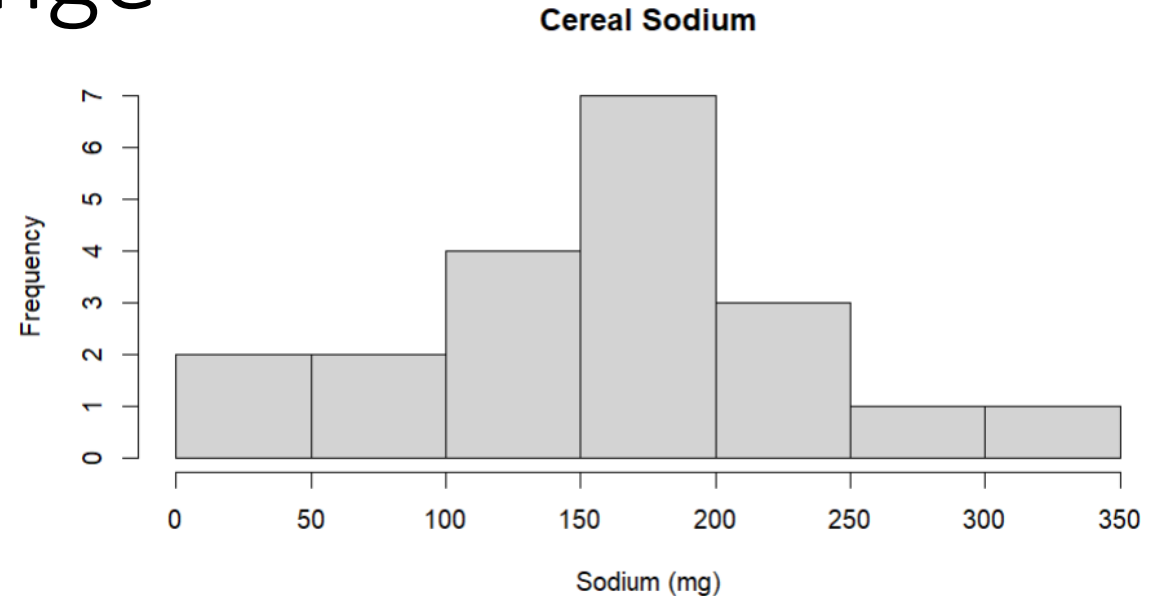


Measures of Spread: Range

- The **range** is a measure of the distance between the smallest and largest values in the data

The range can be computed with only two data points the minimum value and maximum value

- If the range of a set of data is large, then usually this indicates greater dispersion of values
- The range is severely affected by the presence of outliers
- We typically do not use the range to measure variability



Measures of Spread: Interquartile Range

- The **interquartile range (IQR)** measures the spread of the middle 50% of the observations
- It is resistant to outliers

$$IQR = Q3 - Q1$$

- The more variability the larger the value of the **IQR**
- **IQR** is a good choice for distributions that are highly skewed!

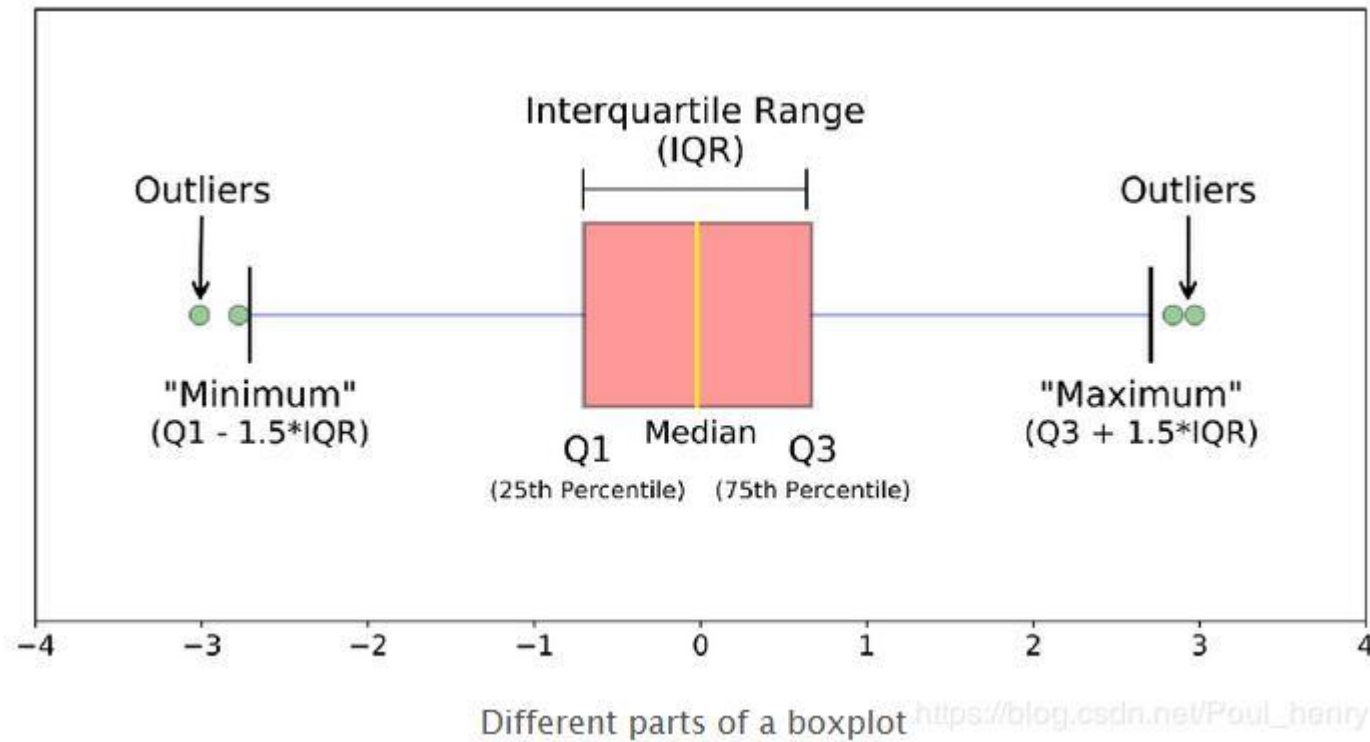
Practice: Finding quartiles and IQR

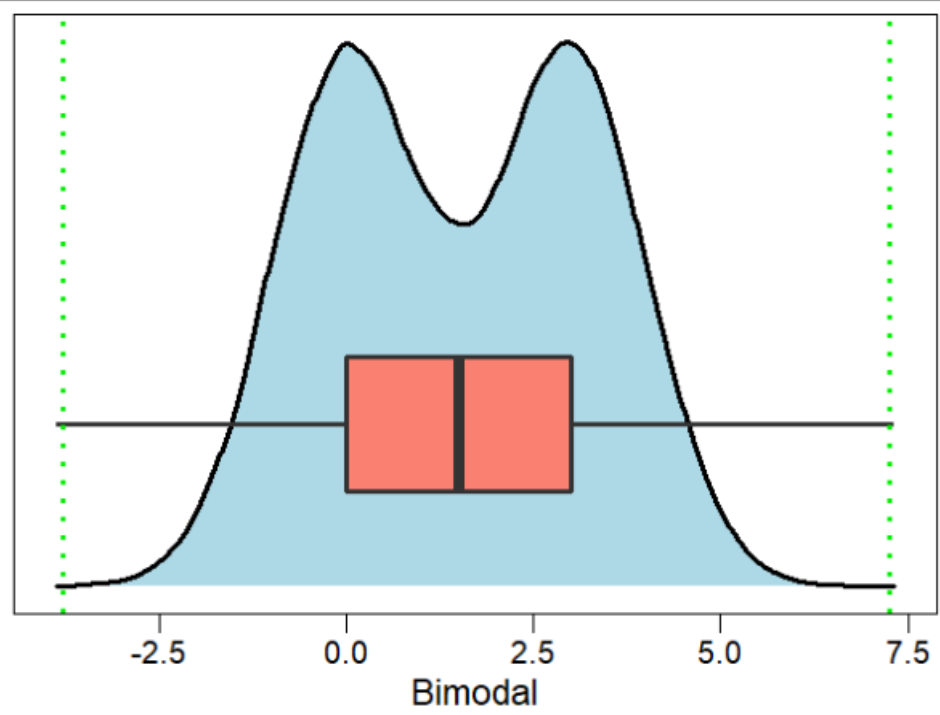
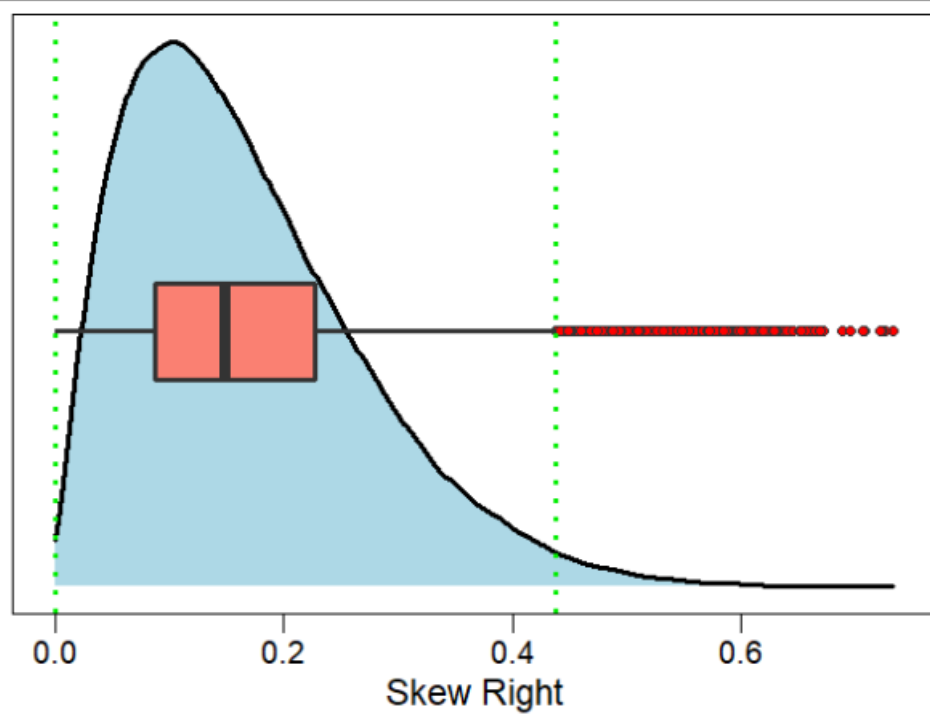
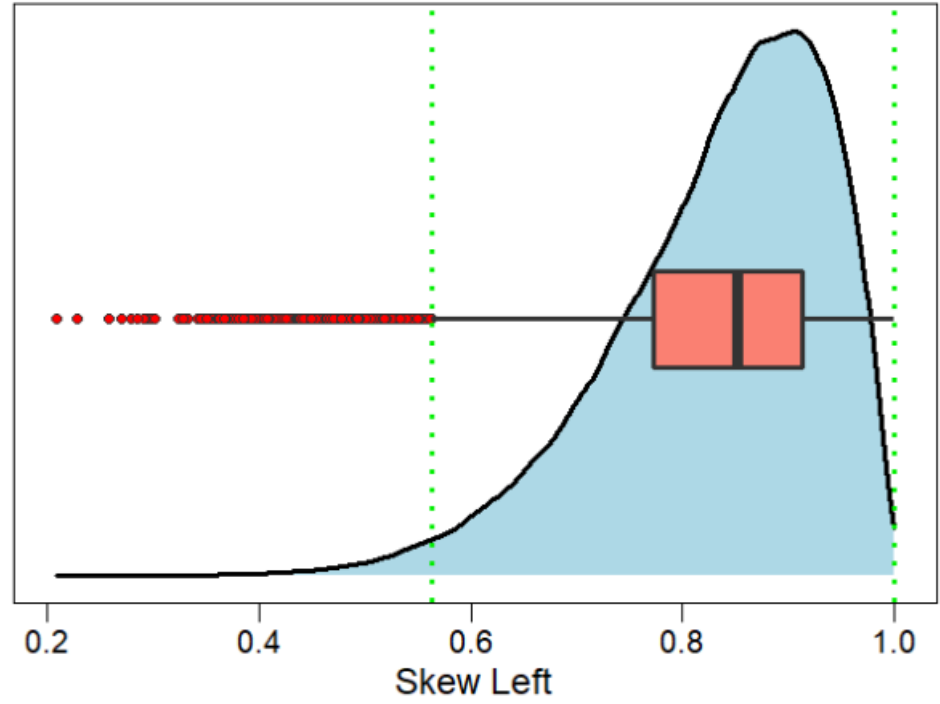
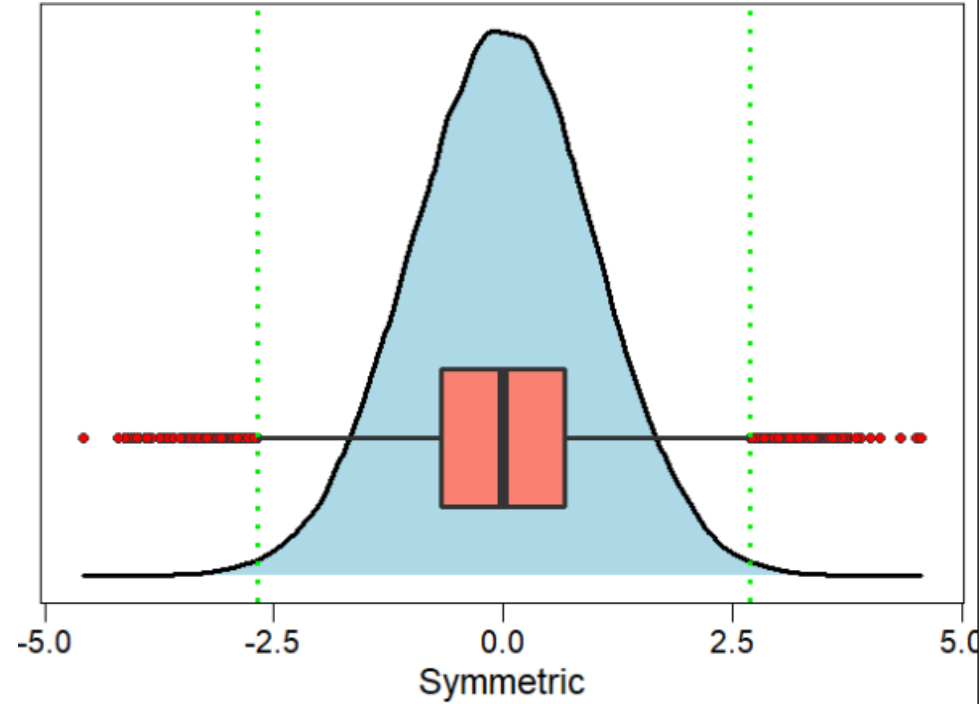
Exam Scores 61,61,65,65,66,68,69,73,74,75,76,78,79,90,94

Compute the IQR

The Boxplot (Box and Whisker Plot): A five number summary

- Pros:
 - good for describing shape and location
 - Can be used to identify outliers
 - Length of whiskers indicates skew
 - Good for comparing two distributions or across categories
- Cons:
 - does not show certain features like mounds, or gaps as well as a histogram





Ex. Construct a Boxplot

- Consider the following data which come from 20 rolls of a six-sided die

• Data =

lower half	middle	upper half
1, 1, 1, 2, 2, 2, 2, 2	2, 3	3, 3, 3, 3, 4, 5, 5, 6, 6, 6

$$Q2 = \text{median} = \frac{2+3}{2} = 2.5$$

$$Q1 = \text{median lower half} = \frac{2+2}{2} = 2$$

$$Q3 = \text{median upper half} = \frac{4+5}{2} = 4.5$$

Practice

Construct the boxplot for student exam scores

Exam Scores 61,61,65,65,66,68,69,73,74,75,76,78,79,90,94

Ex2. Construct a Boxplot

Consider the following 12 observations of a quantitative variable X

$X = \{ -5.7, -2.6, -1.5, -1.3, -0.4, 0.2, 1.5, 2.2, 2.3, 2.6, 2.9, 10.4 \}$

Compute the 5 number summary and draw a boxplot

Compare the IQR with the range

Measures of Spread: Deviation

- A better measure of variability that uses *all* the data is based on **deviations**
- **deviations** are the distances of each value from the mean of the data:

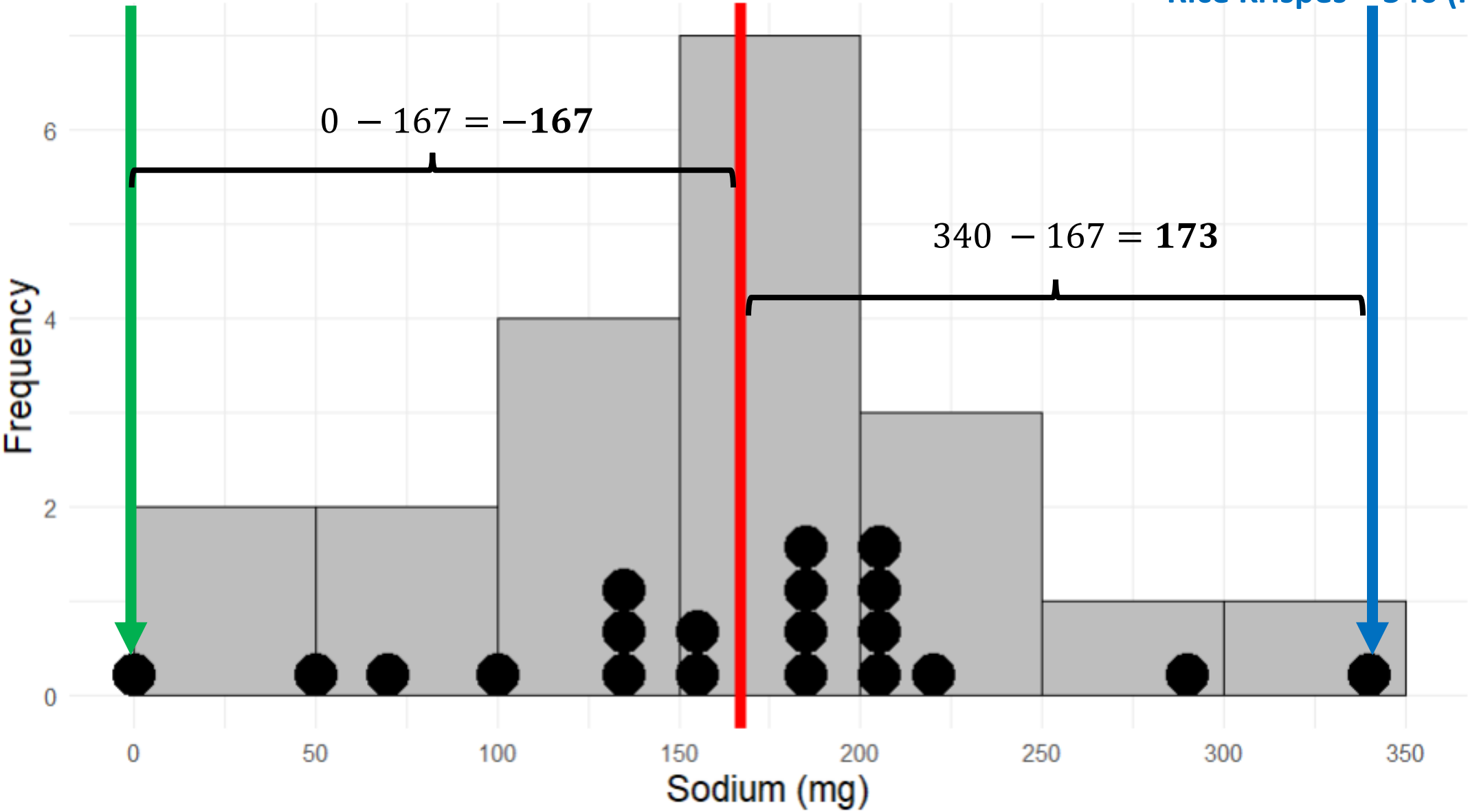
Deviation of an observation $x_i = (x_i - \bar{x})$

- Every observation will have a deviation from the mean

Frosted Mini Wheats = 0 (mg)

Mean = 167 (mg)

Rice Krispes = 340 (mg)



Measures of Spread: Variance

- The sum of all deviations is zero. $\sum_{i=1}^n (x_i - \bar{x}) = 0$
- We typically use either the **squared deviations** or their **absolute value**
Squared deviation of an observation $x_i = (x_i - \bar{x})^2$
- The **Variance** of a distribution is the average squared deviation from the mean

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- The sum $\sum_{i=1}^n (x_i - \bar{x})^2$ is called the sum of squares

Measures of Spread: Standard Deviation

- Since the variance uses the squared deviation, we usually take its square root called the **standard deviation**

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- The standard deviation represents (roughly) the average distance of an observation from the mean
- The greater s is the greater the variability in the data is
- We denote the population parameter for the variance and standard deviation using σ for s and σ^2 for s^2

Why divide by $n - 1$?

- We divide by $n - 1$ because we have only $n - 1$ pieces of independent information for s^2
- Since the sum of the deviations must add to zero, then if we know the first $n - 1$ deviations we can always figure out the last one
- Ex.) suppose we have two data points and deviation of the first data point is $x - \bar{x} = -5$
 - Then the deviation of the second data point has to be 5 for the sum of deviations to be zero.

Try it out: Computing s and s^2

- Roll a six-sided die $n = 10$ times and record the number rolled each time
- Data = 1,2,3,3,4,4,4,5,6,6
- Mean = 3.8

